



Peter Schneider-Kamp

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Co-lead of Danish Foundation Models

Dept. Mathematics & Computer Science

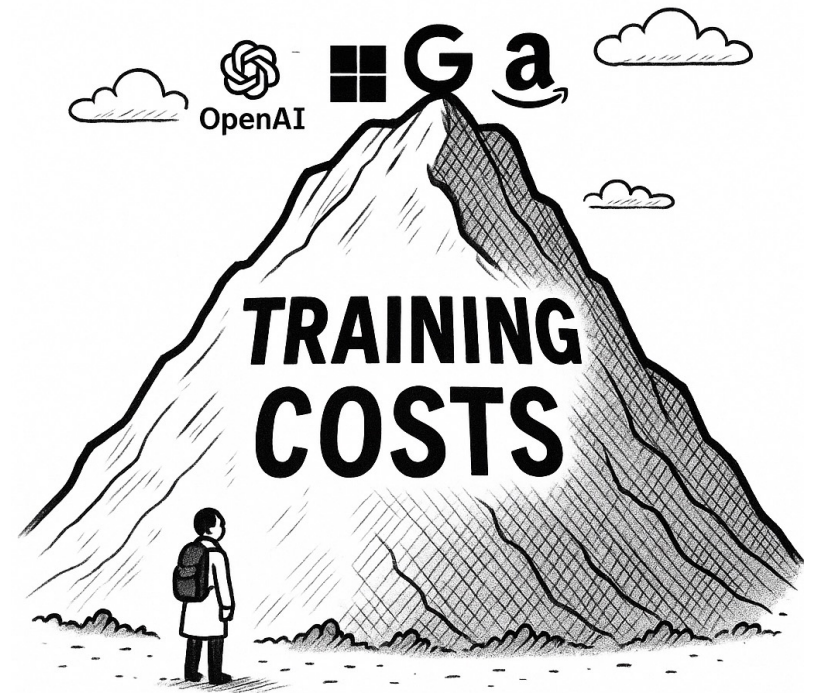
University of Southern Denmark

**The cost of
intelligence:**

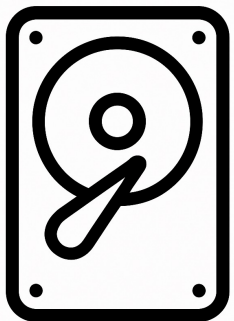
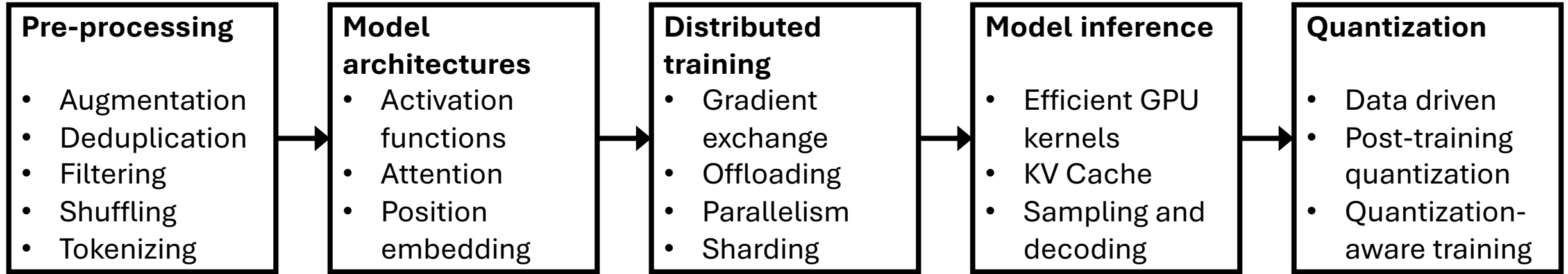
**Efficiency is the
only path to
democratized AI**

From open dreams to closed reality

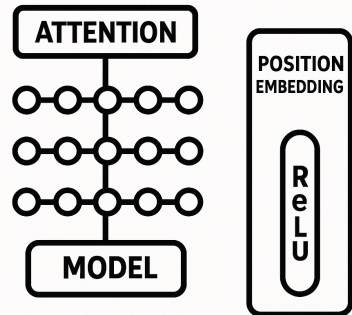
- Training frontier models costs tens of \$millions
- Requires large investments in GPU/TPU infrastructure
- Academia, startups, & Global South effectively priced out



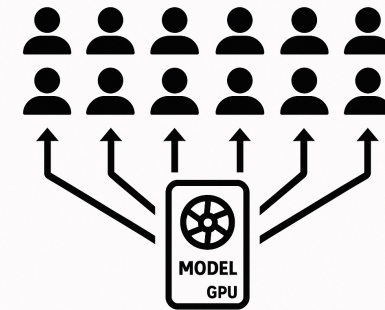
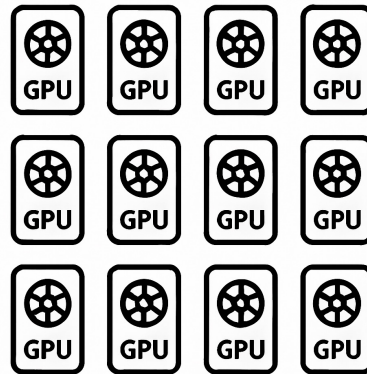
From raw data to deployed model



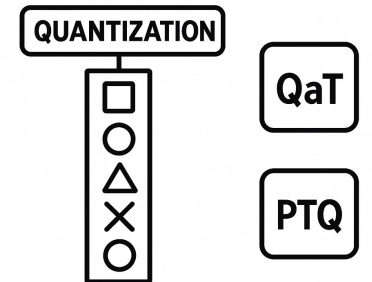
**TEXT
DATA
8TB**



MODEL ARCHITECTURES



THOUSANDS OF USERS

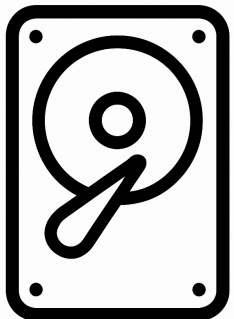


QUANTIZATION

From raw data to deployed model

Pre-processing

- Augmentation
- Deduplication
- Filtering
- Shuffling
- Tokenizing

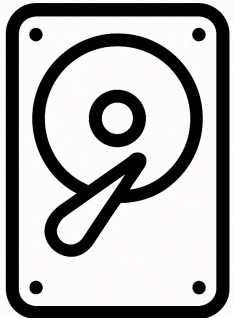


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Challenge:

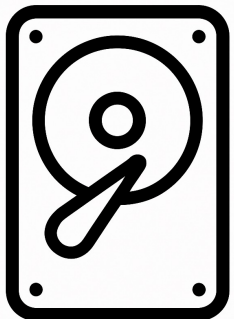
- even text datasets fill multiple TBs of storage
- often limited multi-modal support
- many data formats optimized for streaming
- global in-memory operations do not scale

How to globally shuffle common pile?

From raw data to deployed model

Pre-processing

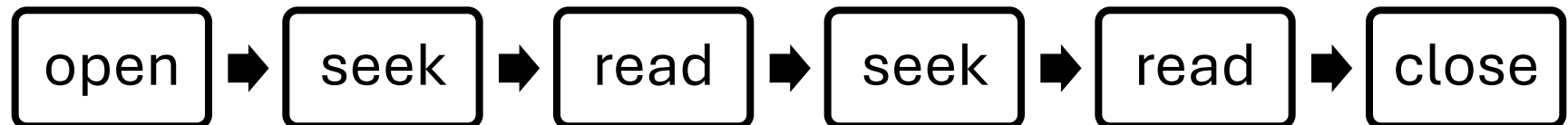
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TEXT
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Improving existing data format (MDS):

- MDS reference implementation is slow
- bulk reader for full-speed sequential reads
- random access reader with full index usage
- sample- instead of shard-level compression



From raw data to deployed model

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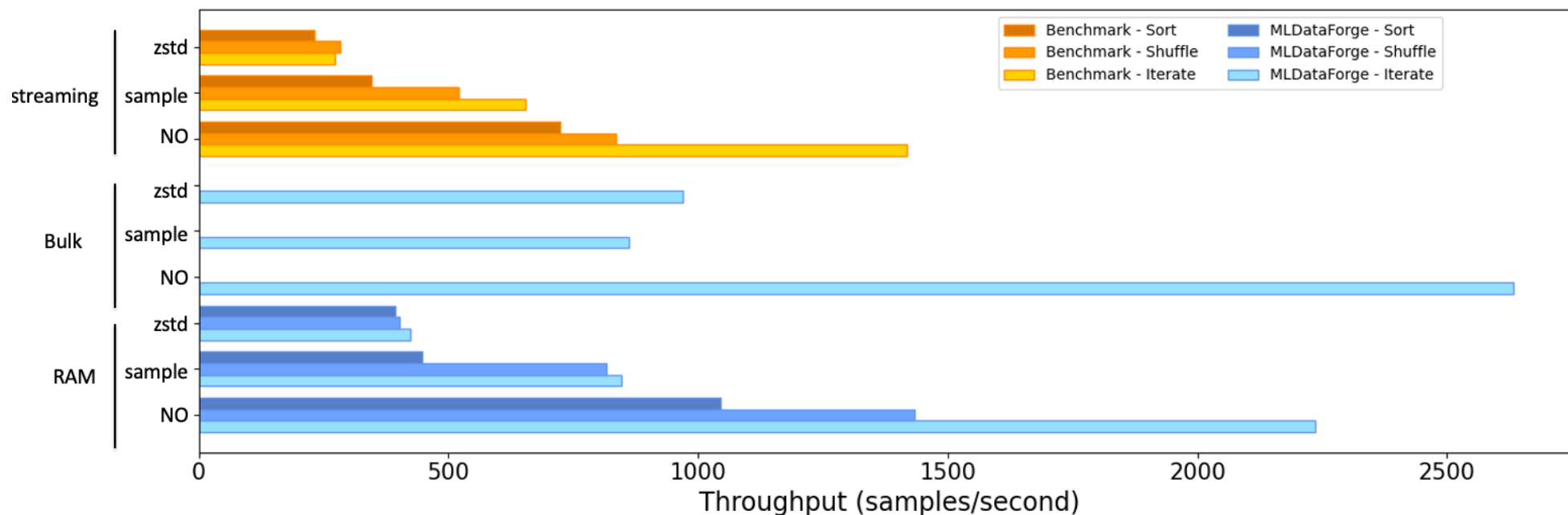
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TEXT
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MDS throughput

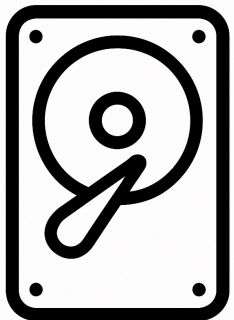
[RANLP 2025]



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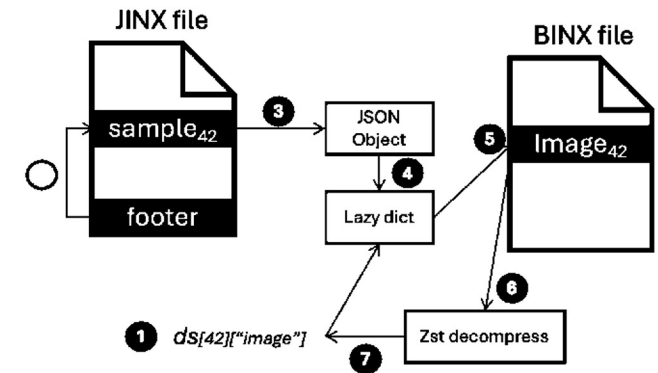
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Innovate new data format (JINX):

- combining the best of JSONL and MDS
- human-readable index and meta data
- encoded and sidecared (BINX) binary
- lazy decoding and sidecaring



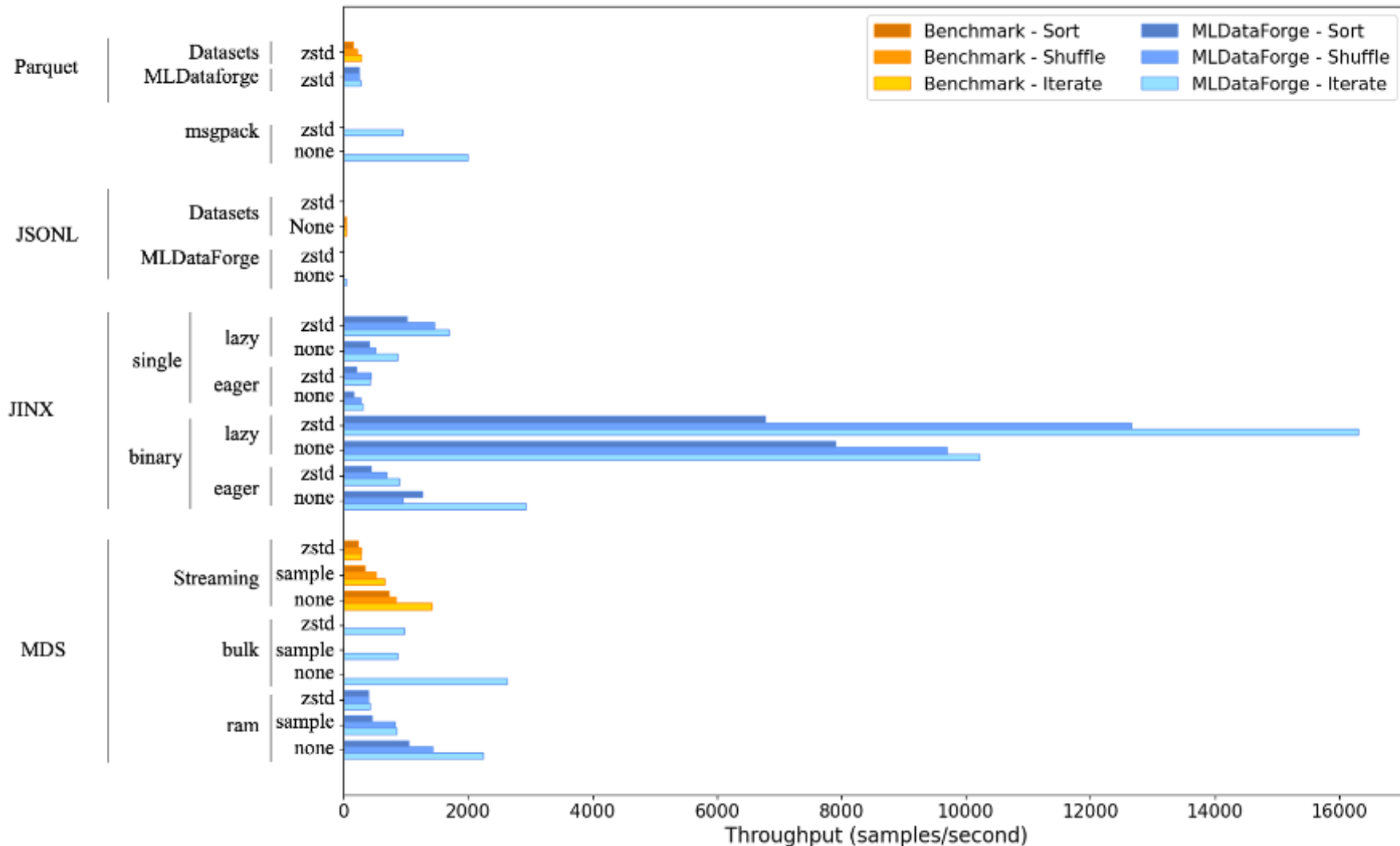
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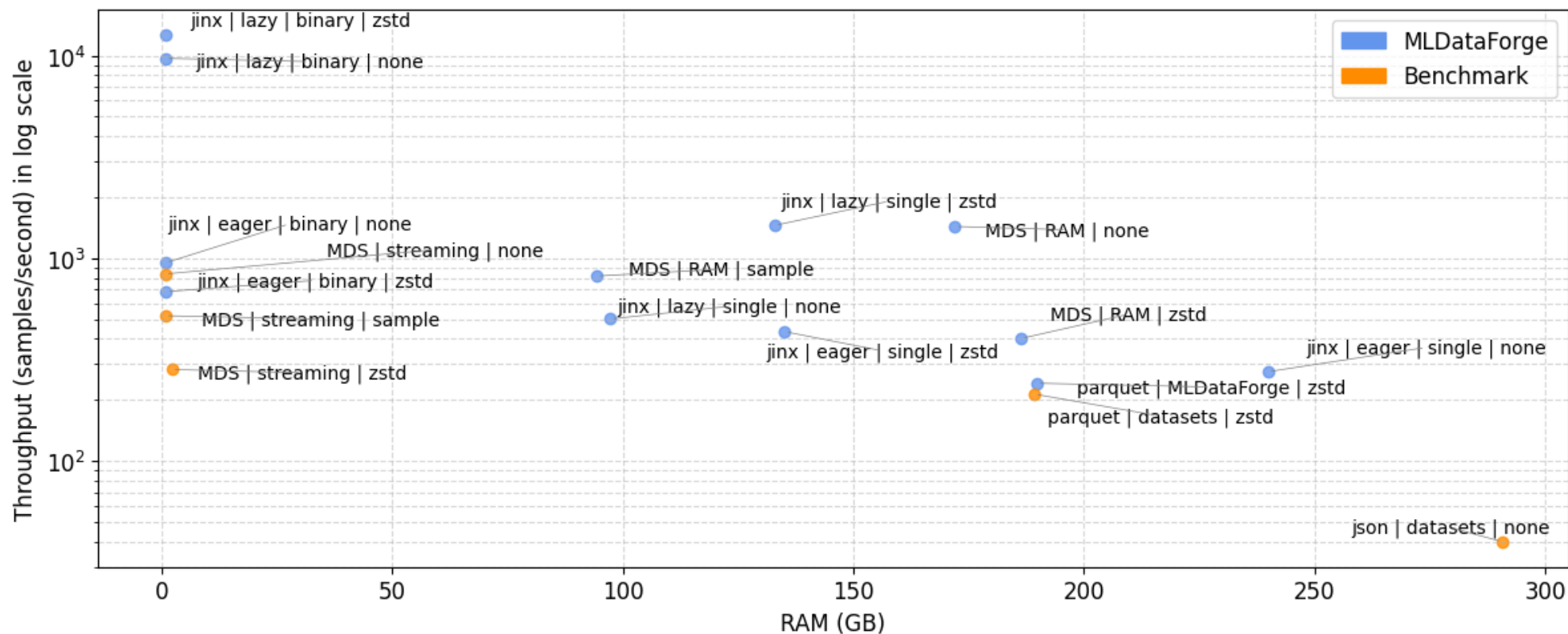
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Throughput vs RAM

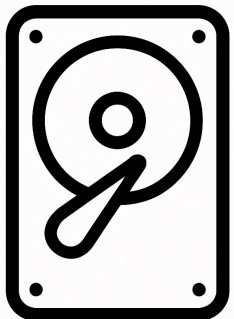
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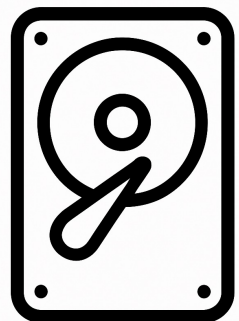
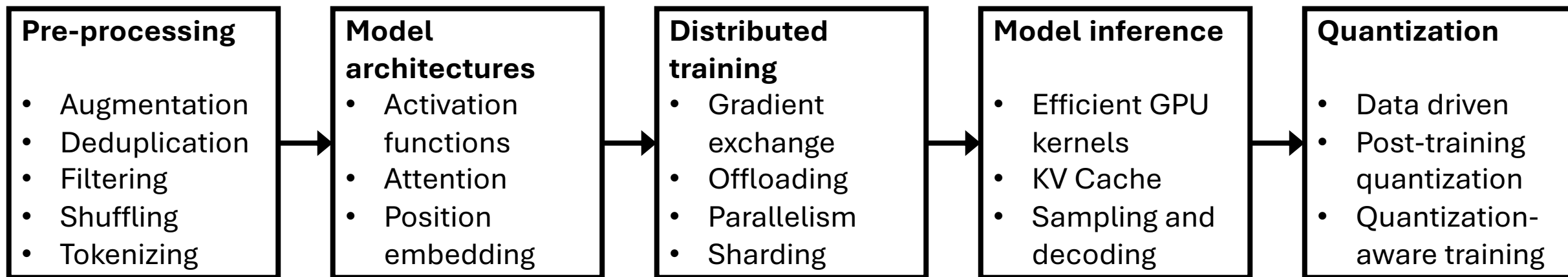
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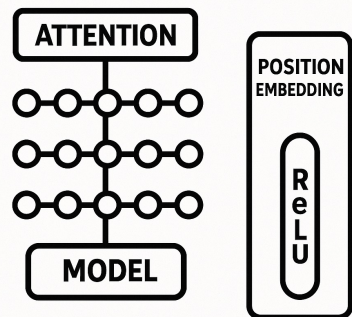


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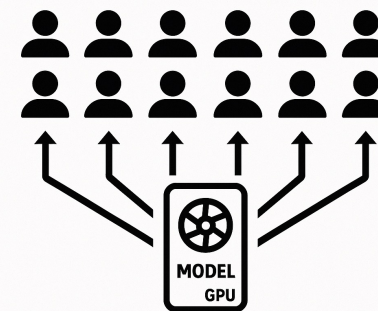
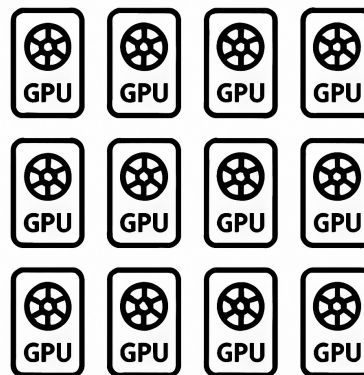
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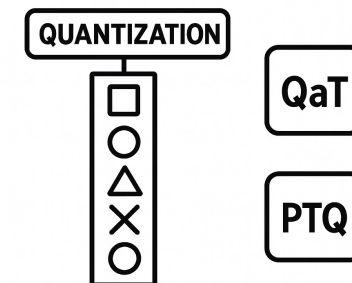
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MODEL ARCHITECTURES



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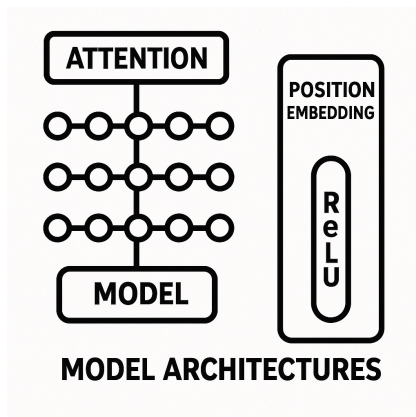


QUANTIZATION

From raw data to deployed model

Model architectures

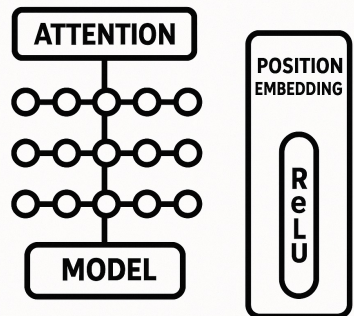
- Activation functions
- Attention
- Position embedding



From raw data to deployed model

Model architectures

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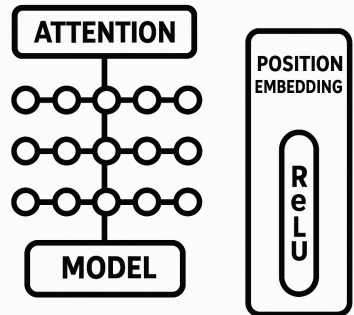
Revisiting GPT-2 (NanoGPT):

- GPT-2 training took weeks on many TPUs
- trained on 40 GB WebText ($\approx 20\text{B}$ tokens)
- NanoGPT took 4 days on 8xA100 40GB
- trained on 40 GB OpenWebText

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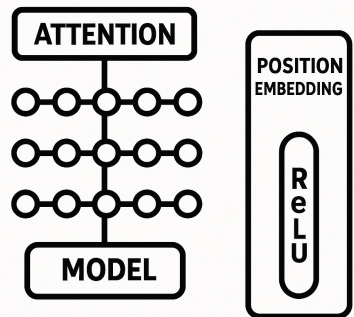
Efficiency potential (NanoGPT speedrun):

- llm.c took 45 min on 8xA100 40GB
- trained on 10B tokens from FineWeb
- modded-nanogpt 3 min on 8xH100 80GB
- Trained on 0.7B tokens from FineWeb

From raw data to deployed model

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- Activation functions
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MODEL ARCHITECTURES

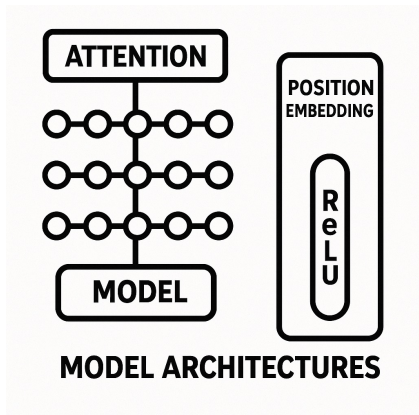
Modern architecture (modded-nanogpt):

- RoPE, QKNorm, ReLU², Muon optimizer
- FP8 for head, softcap logits (á la Gemma 2)
- limited connectivity (á la U-Net)
- FlexAttention (sliding window & warmup)

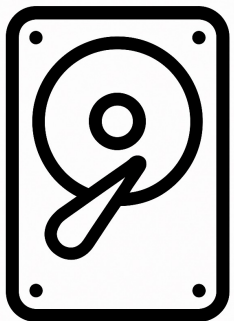
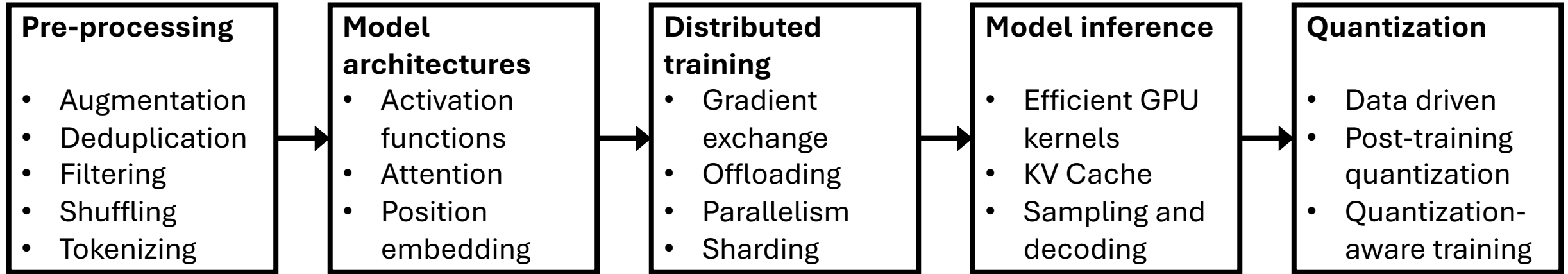
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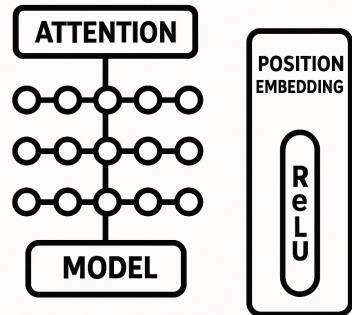
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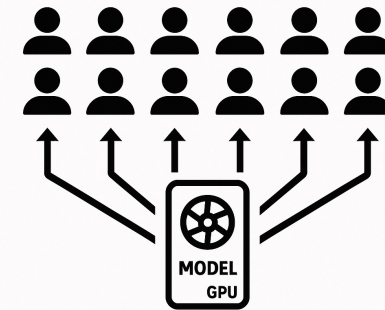
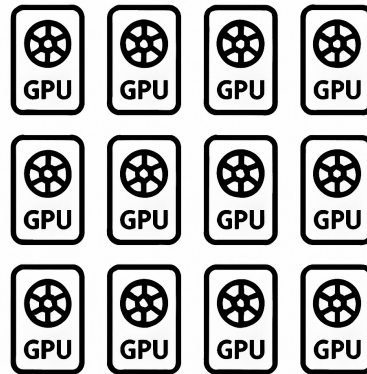
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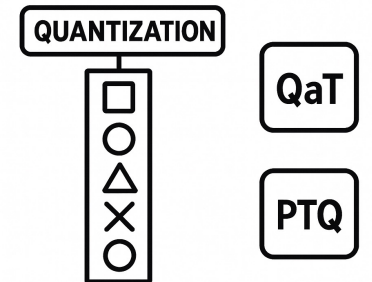
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MODEL ARCHITECTURES



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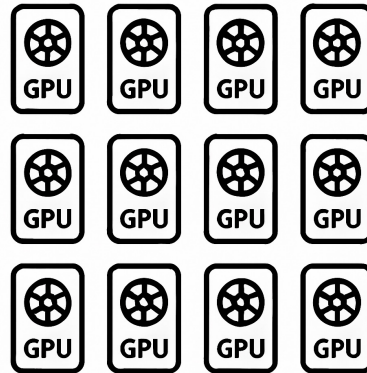


QUANTIZATION

From raw data to deployed model

Distributed training

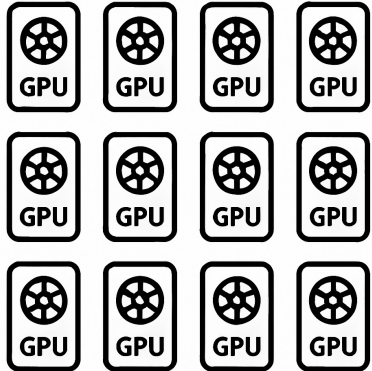
- Gradient exchange
- Offloading
- Parallelism
- Sharding



From raw data to deployed model

Distributed training

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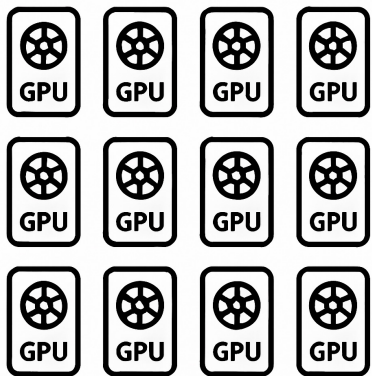
Challenge:

- scaling to multiple GPUs incurs overhead
- intra-node BW of 63 –1800 GB/s
- inter-node BW 0.125 –100 GB/s
- gradient synchronization dominates soon

From raw data to deployed model

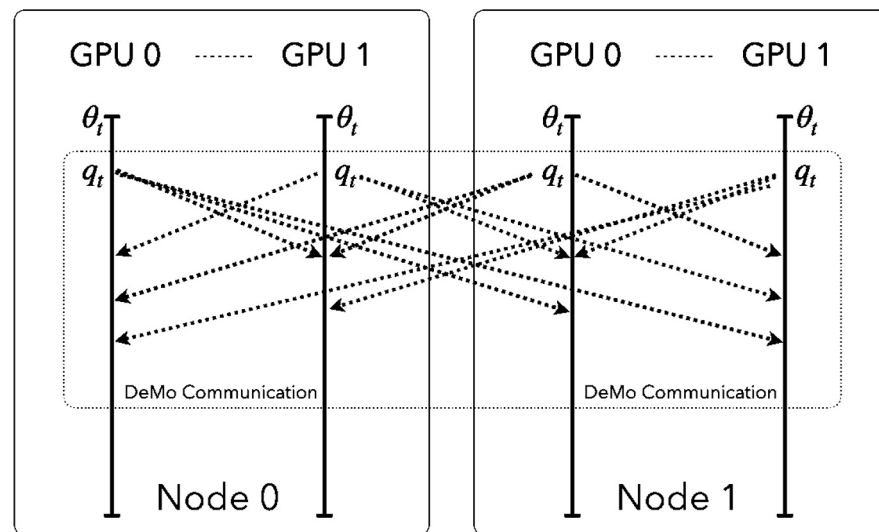
Distributed training

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Decoupled Momentum (NousResearch):

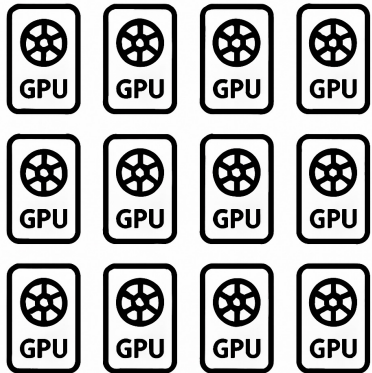
- distributed data-parallel training
- discrete cosine transform to identify the fastest-moving gradient components



From raw data to deployed model

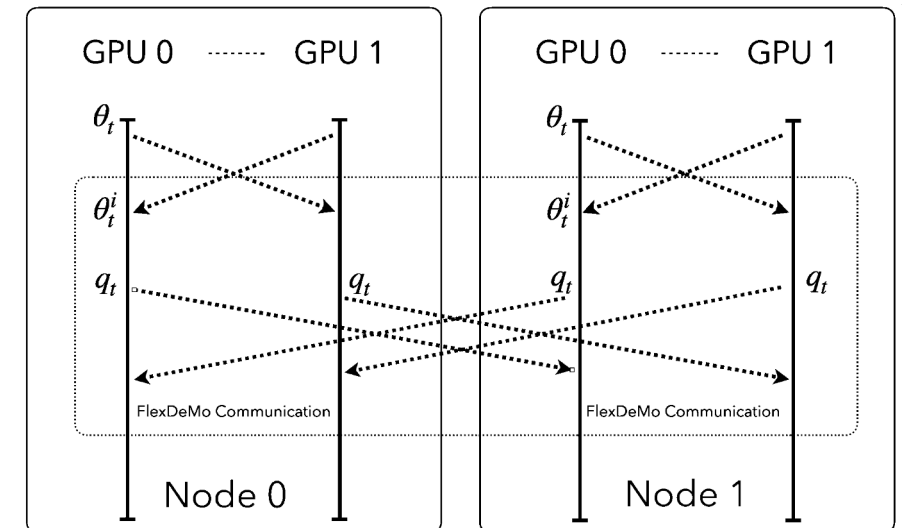
Distributed training

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FlexDeMo [submitted to AAAI]:

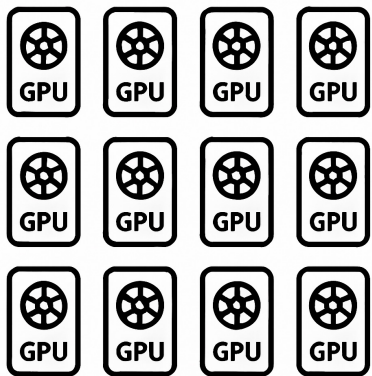
- fully sharded data parallel training
- allows larger models
- scales with #nodes rather than #GPUs



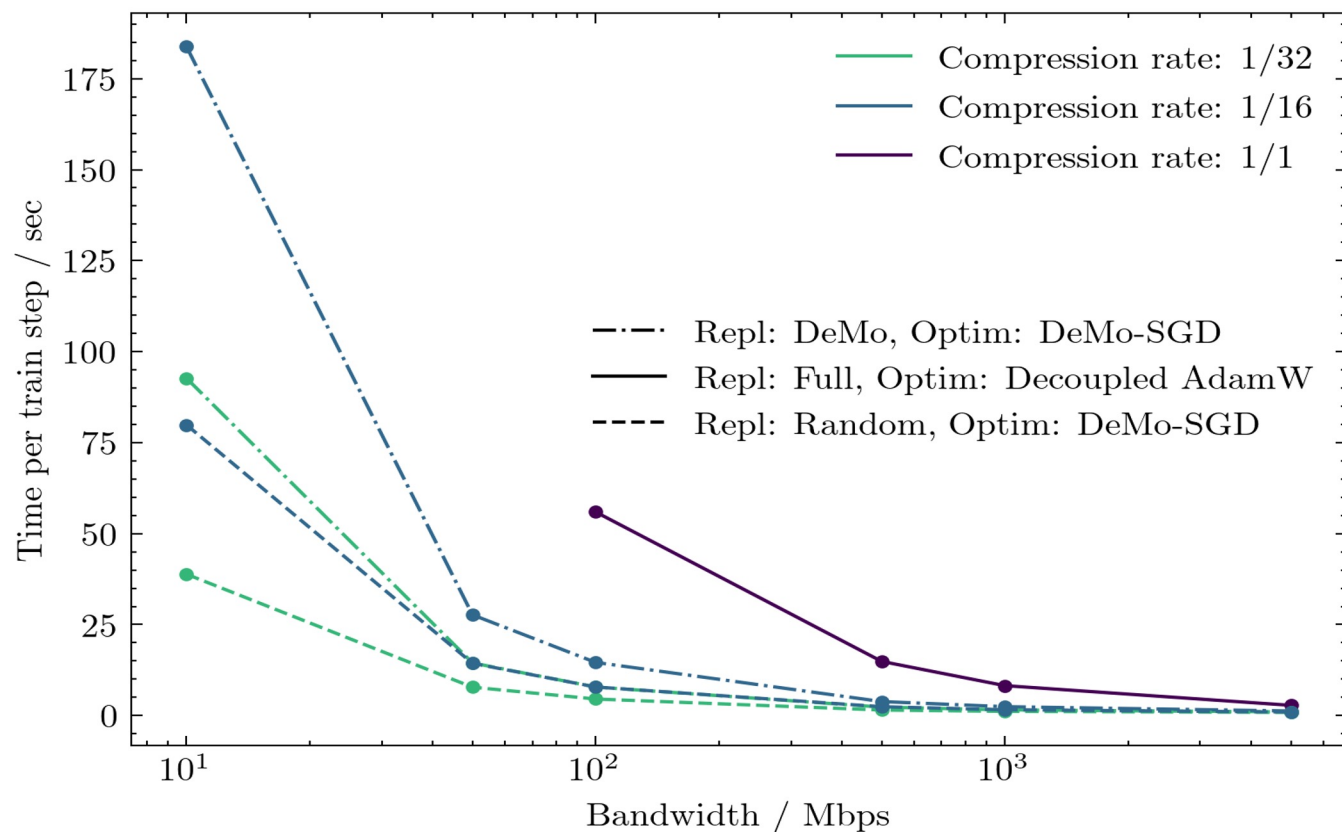
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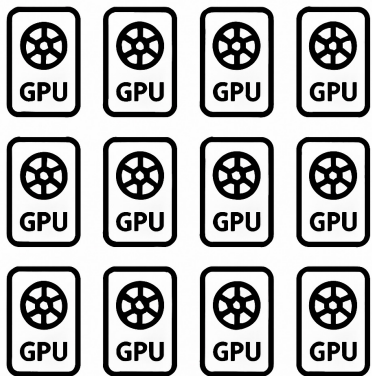
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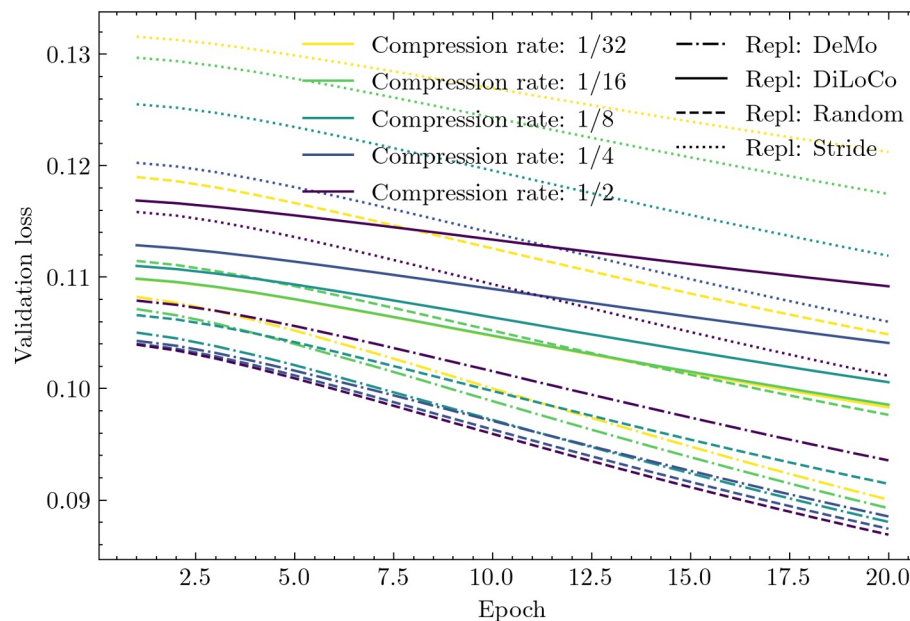
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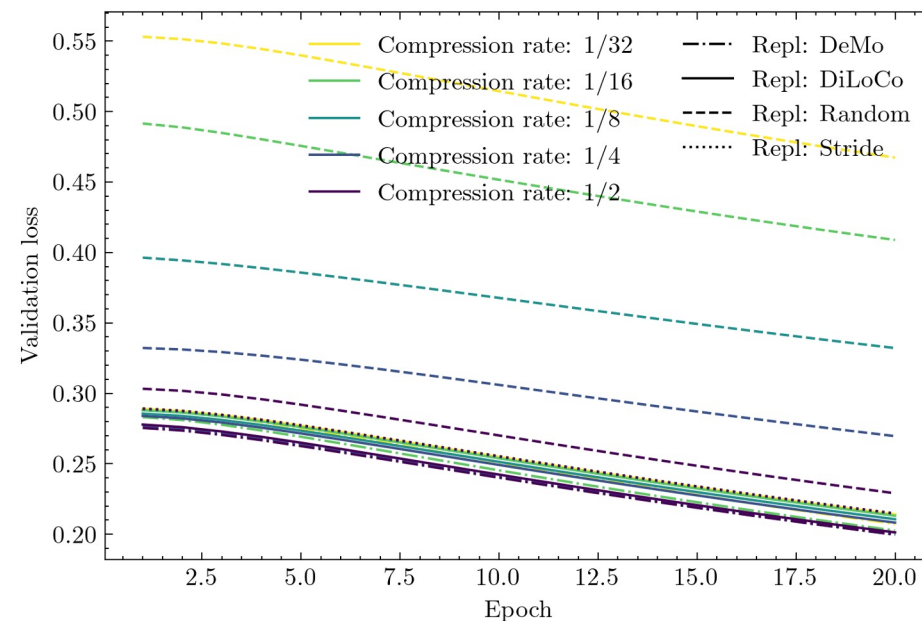
FlexDeMo [submitted to AAAI]:

- random indexing instead of DeMo

T5-Large validation loss on OpusBooks En-Fr



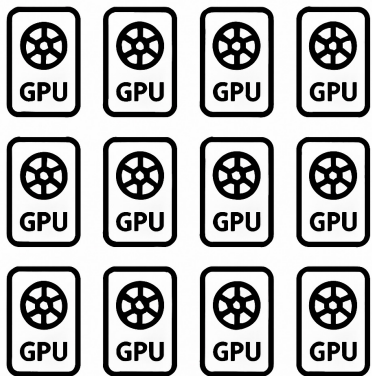
Vit-B Validation loss on Cifar100



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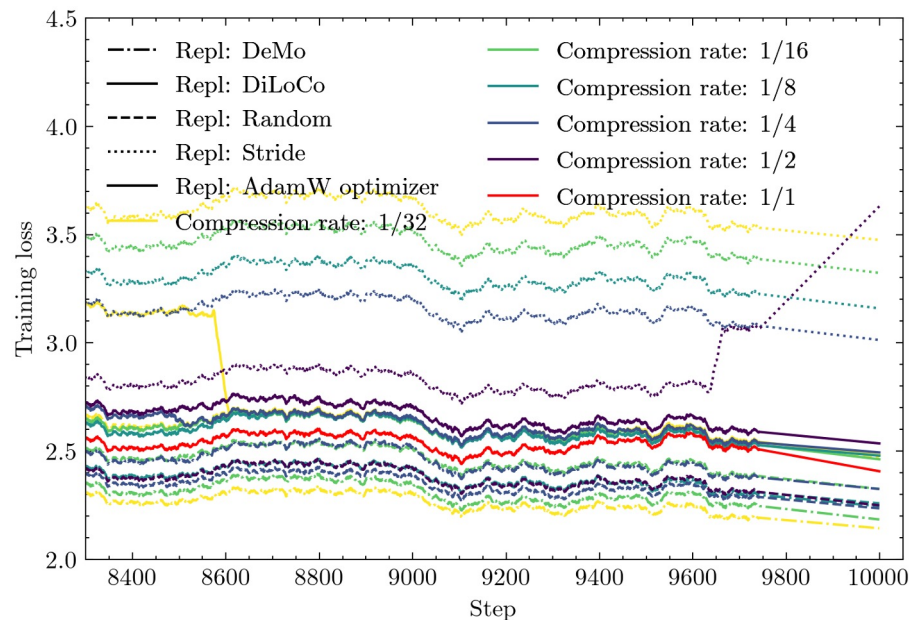
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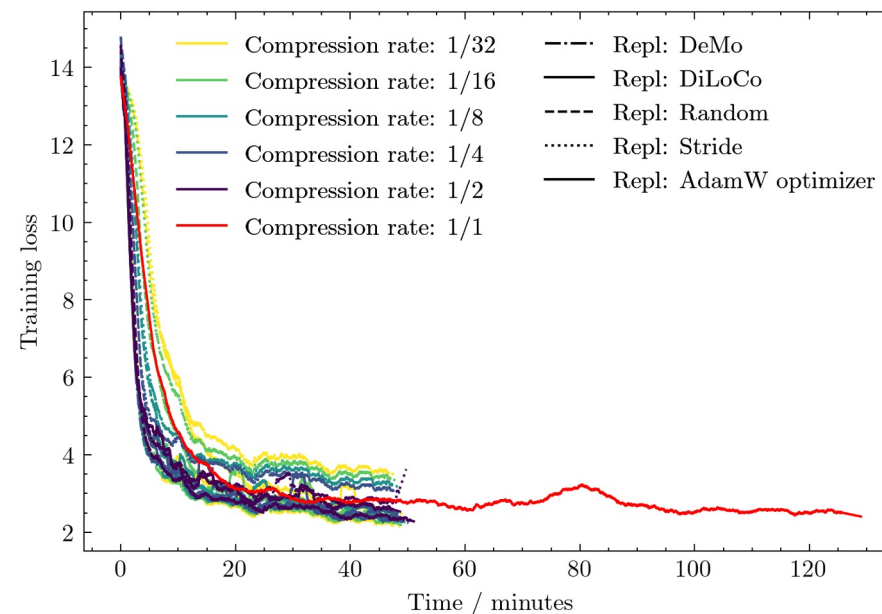
FlexDeMo [submitted to AAAI]:

- training OLMo 1B on 64 nodes (256 GPUs)

OLMo2 1B train loss over 10k training steps



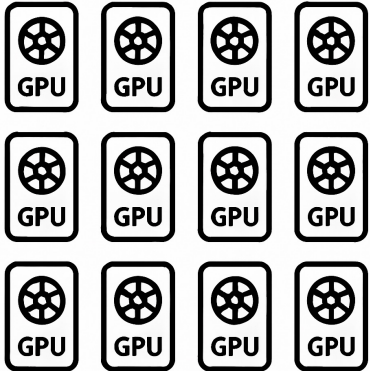
OLMo2 1B train loss per wall time



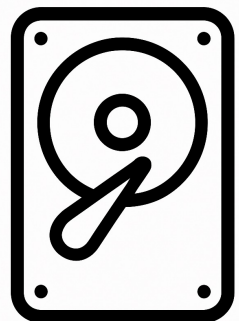
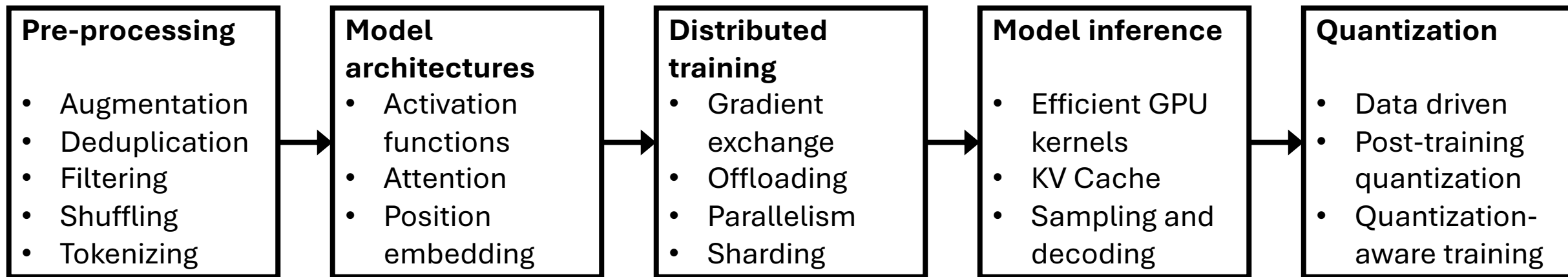
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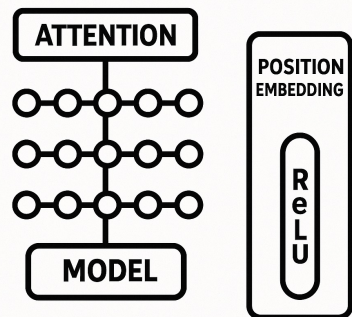
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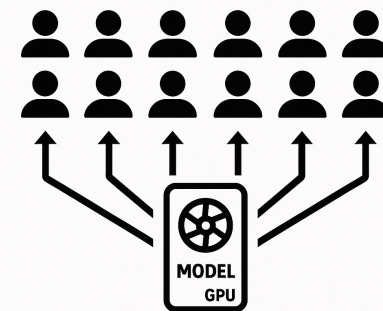
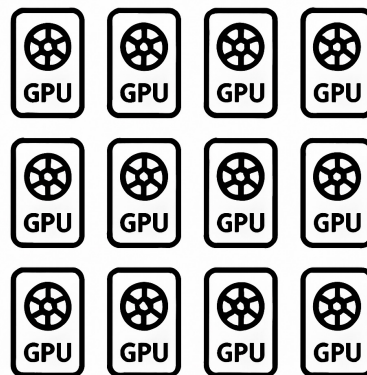
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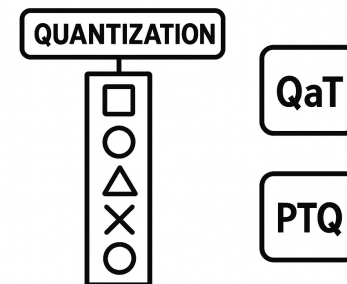
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MODEL ARCHITECTURES



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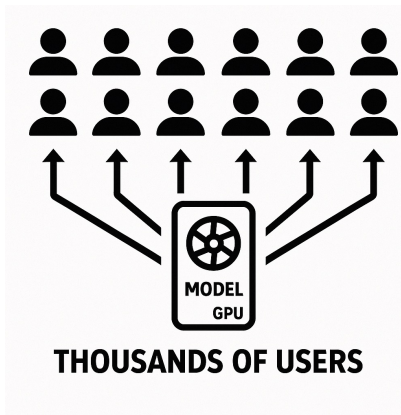


QUANTIZATION

From raw data to deployed model

Model inference

- Efficient GPU kernels
- KV Cache
- Sampling and decoding



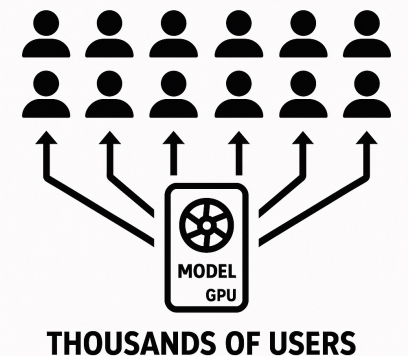
From raw data to deployed model

Efficiency requirements:

- thousands of simultaneous users
- often real-time requirements (TPS, TTFT)
- aggregated cost dwarves training cost
- test-time compute becoming the norm

Model inference

- Efficient GPU kernels
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From raw data to deployed model

Continuous batching (HF TGI, vLLM):

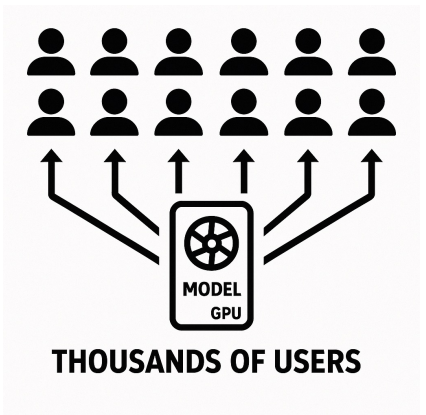
- do not wait for longest sequence
- add new sequences on the fly

Model inference

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- KV Cache
- Sampling and decoding

T_1	T_2	T_3	T_4	T_5	T_6	T_7	T_8
S_1	S_1	S_1	S_1				
S_2	S_2	S_2					
S_3	S_3	S_3					
S_4	S_4	S_4	S_4				

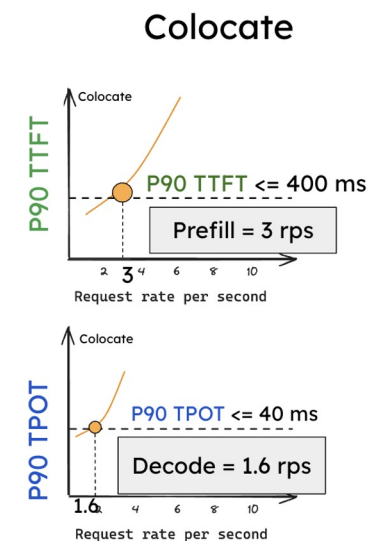
T_1	T_2	T_3	T_4	T_5	T_6	T_7	T_8
S_1	S_1	S_1	S_1	S_1	END	S_6	S_6
S_2	S_2	S_2	S_2	S_2	S_2	S_2	END
S_3	S_3	S_3	S_3	END	S_5	S_5	S_5
S_4	S_4	S_4	S_4	S_4	S_4	END	S_7



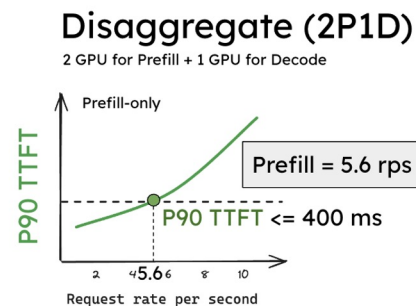
From raw data to deployed model

PD disaggregation (DistServe, SGLang)

- heterogeneity of prefill & decode
- separate GPUs for each phase
- optimize RPS



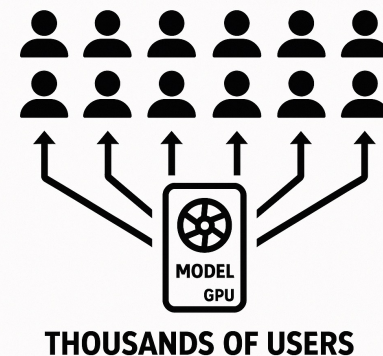
 Max System rps
= $\text{Min}(\text{Prefill}, \text{Decode})$
= 1.6 rps / GPU



   Disaggregate (2P1D)
= $\text{Min}(5.6 \times 2, 10)$ rps / 3 GPU
= 3.3 rps / GPU

Model inference

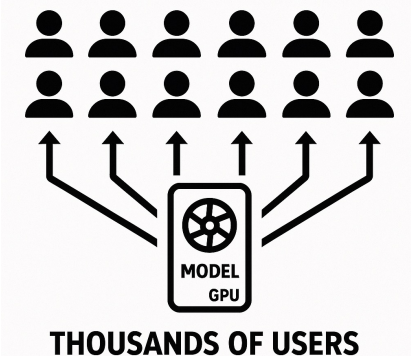
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- KV Cache
- Sampling and decoding



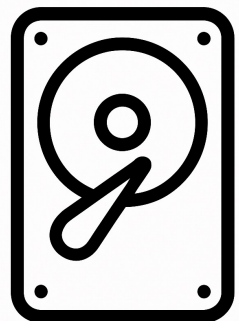
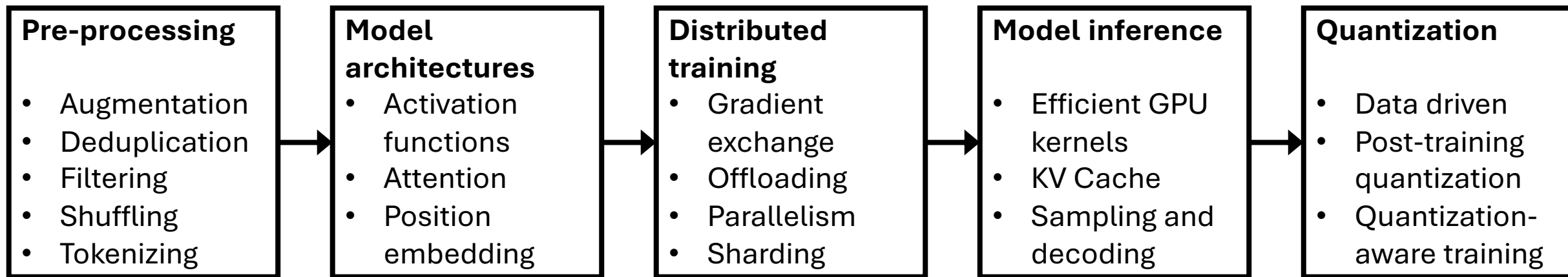
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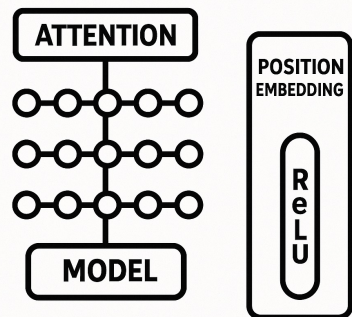
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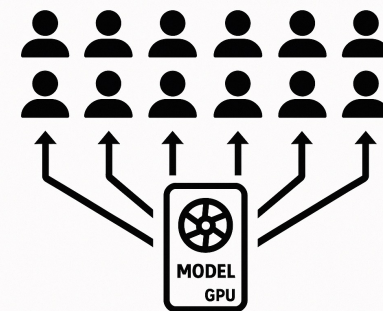
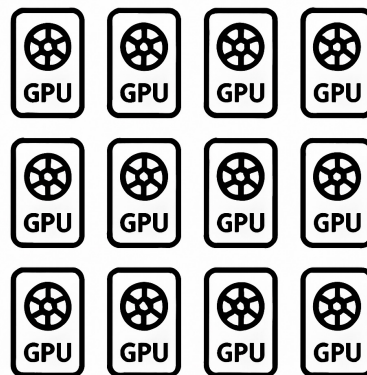
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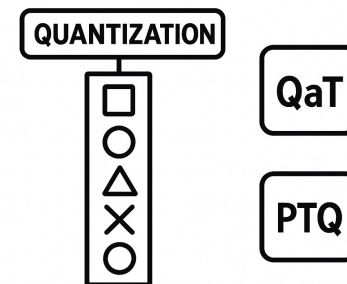
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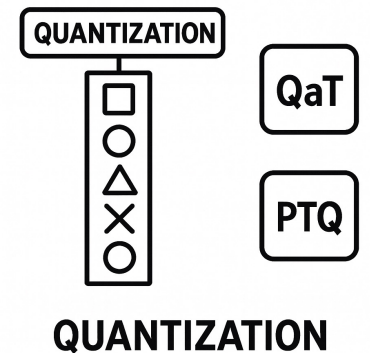


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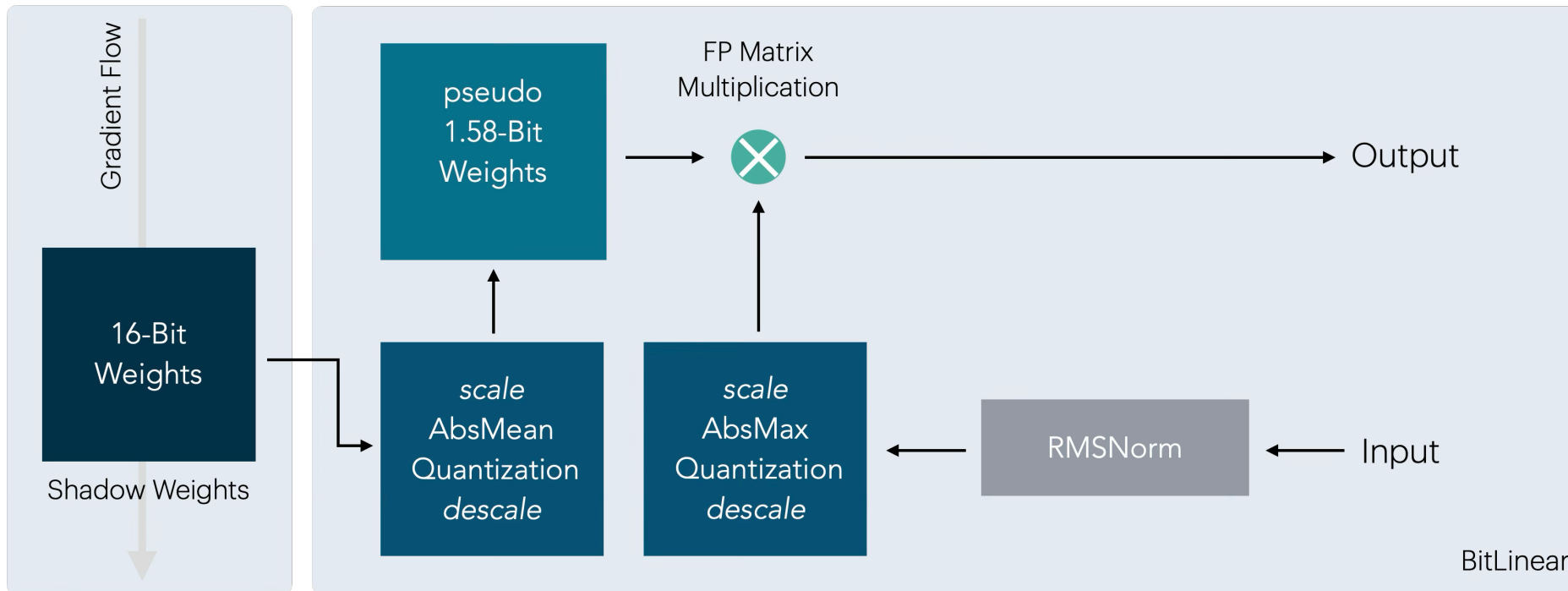
Quantization

- Data driven
- Post-training quantization
- Quantization-aware training



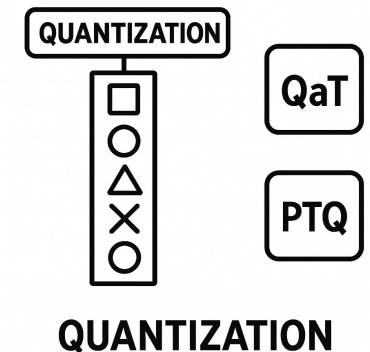
From raw data to deployed model

1.58-bit quantization (MSR):



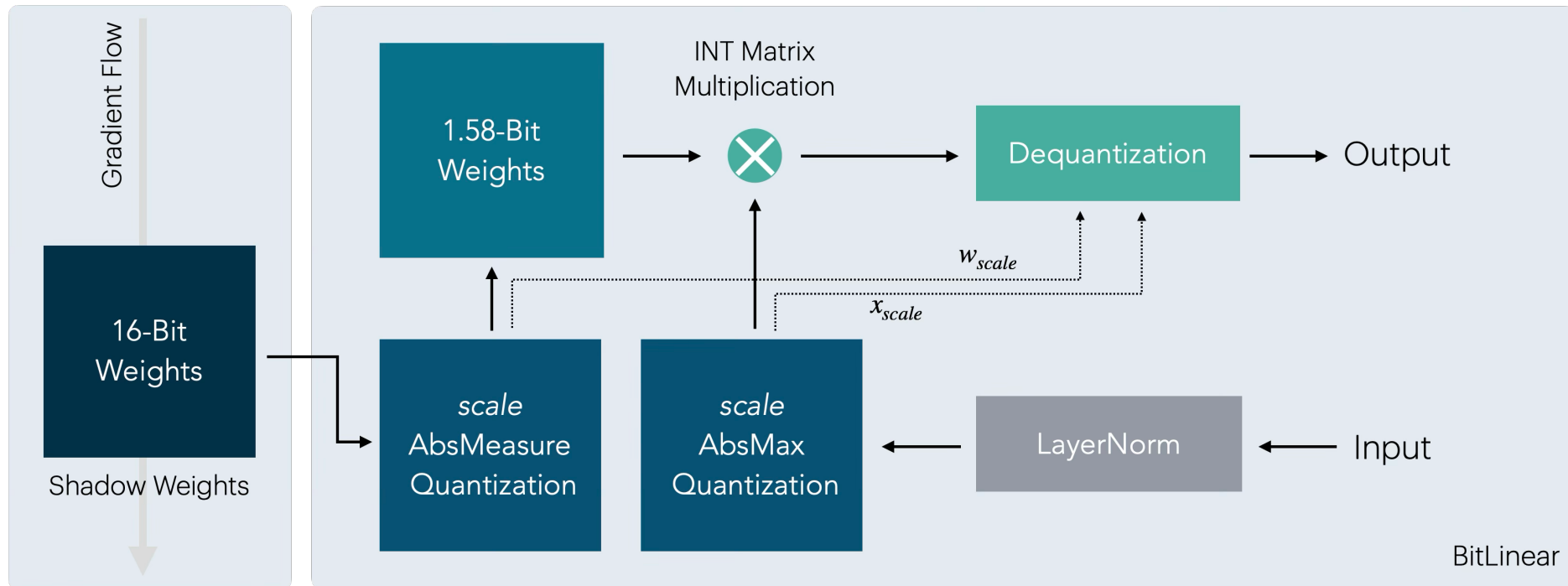
Quantization

- Data driven
- Post-training quantization
- Quantization-aware training



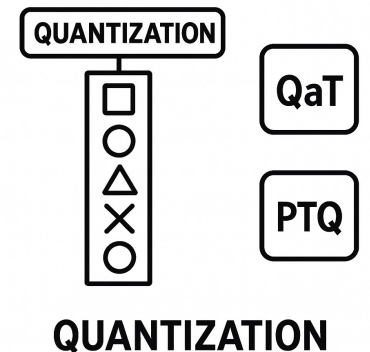
From raw data to deployed model

1.58-bit quantization reloaded [DELTA 2024]:



Quantization

- Data driven
- Post-training quantization
- Quantization-aware training



From raw data to deployed model

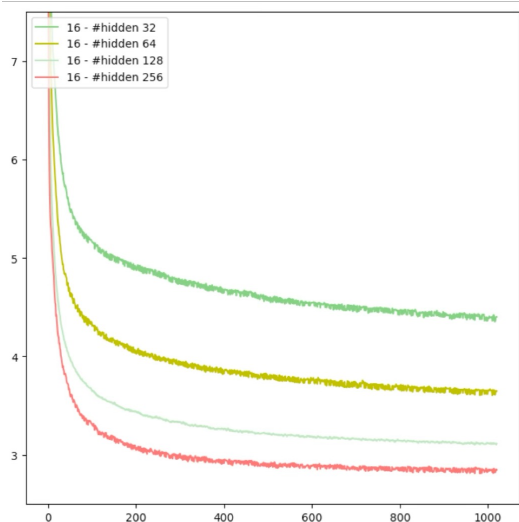
1.58-bit quantization reloaded [DELTA 2024]:

- Scaling law for SLMs: $d_{\text{model}} \rightarrow 2 \times d_{\text{model}}$

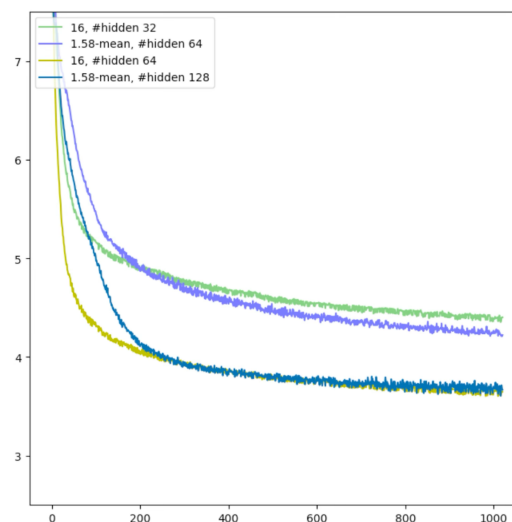
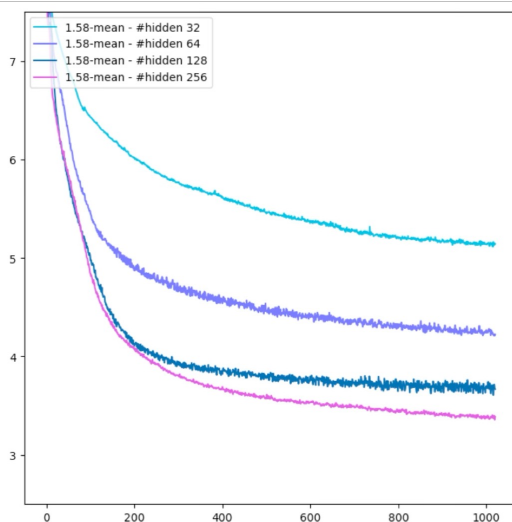
Quantization

- Data driven
- Post-training quantization
- Quantization-aware training

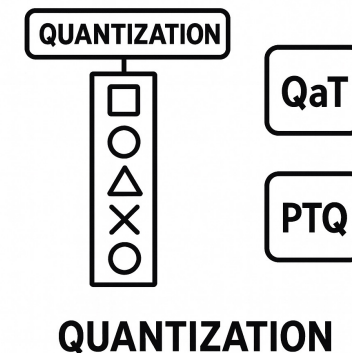
(a) Scaling for 16 bit.



(b) Scaling for 1.58 bit (mean).



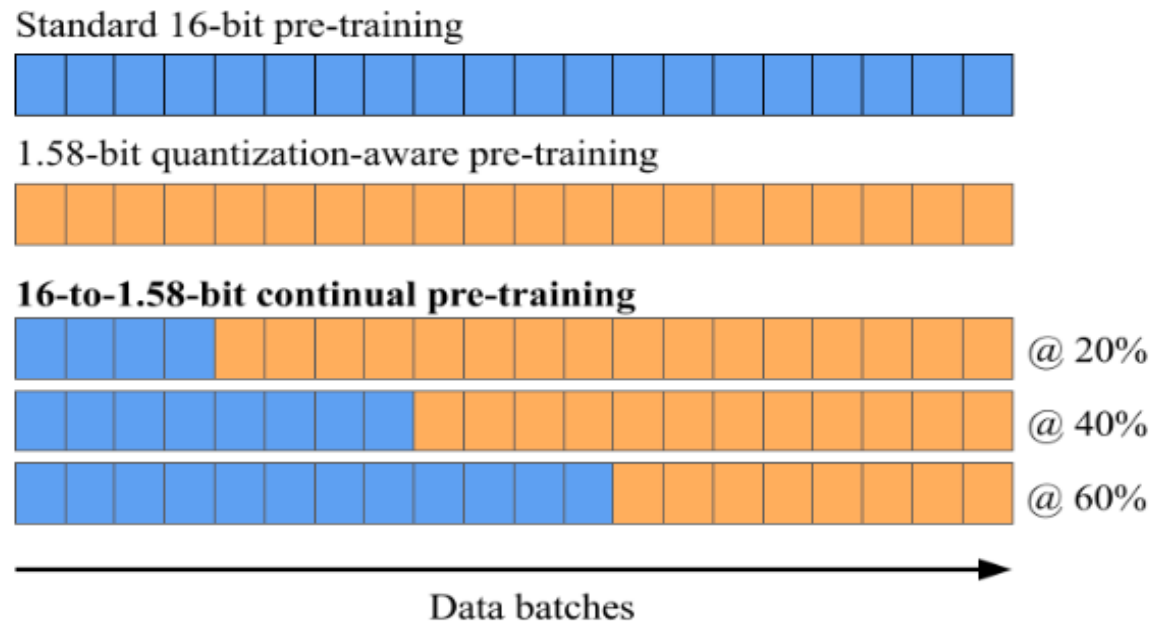
(c) Scaling 16 vs 1.58 bit (mean).



From raw data to deployed model

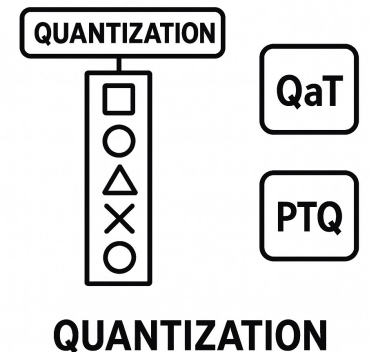
Continual 1.58-bit training [ACL 2025]:

- state-of-the-art is full 1.58-bit training
- transition from 16- to 1.58-bit



Quantization

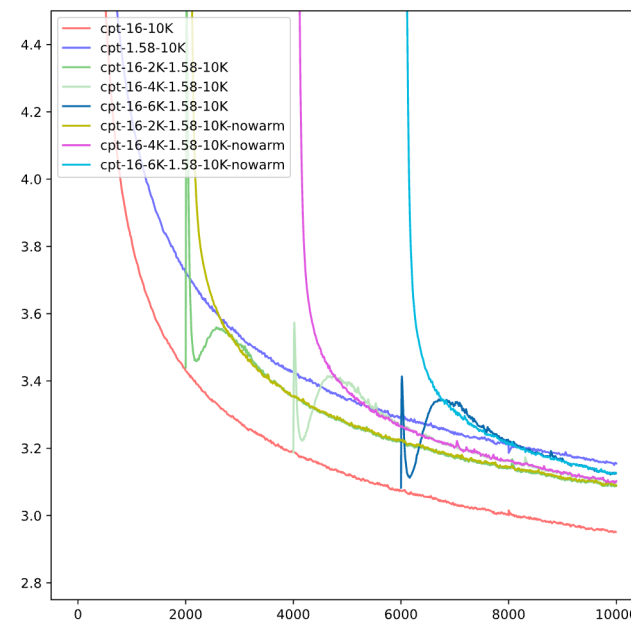
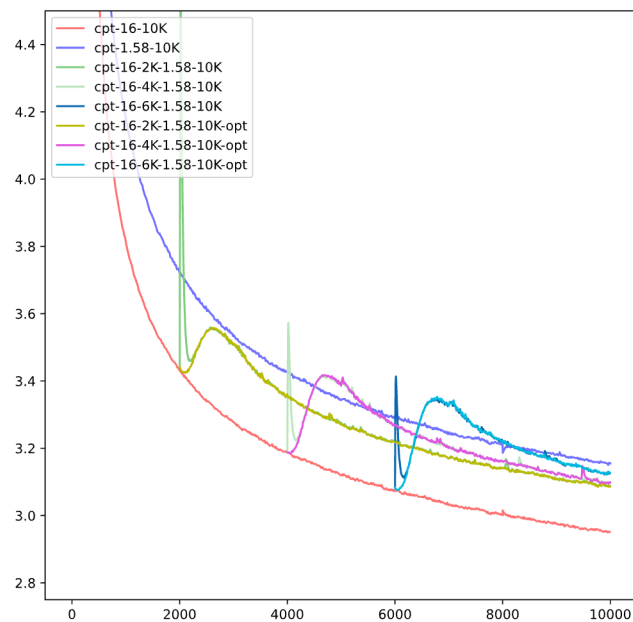
- Data driven
- Post-training quantization
- Quantization-aware training



From raw data to deployed model

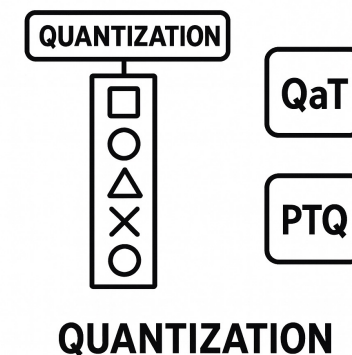
Continual 1.58-bit training [ACL 2025]:

- optimizer state & phase-in are not necessary



Quantization

- Data driven
- Post-training quantization
- Quantization-aware training



From raw data to deployed model

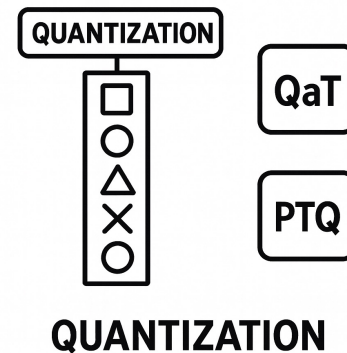
Continual 1.58-bit training [ACL 2025]:

- mixed pre-training beats full 1.58-bit & 16-bit training on 9 of 11 downstream tasks

Quantization

- Data driven
- Post-training quantization
- Quantization-aware training

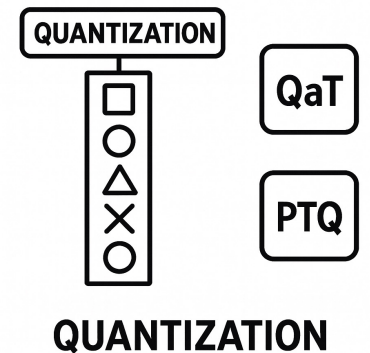
Model	ARC-easy	CommitB	COPA	HellaSwag	MRPC	OpenBookQA	PIQA	RTE	SciQ	SST-2	WinoGrande
full 1.58-bit training	0.4561	0.4107	0.6300	0.3212	0.8122	0.2760	0.6425	0.5343	0.6840	0.5803	0.5185
full 16-bit training	0.4596	0.4107	0.6500	0.3607	0.8122	0.2760	0.6589	0.5379	0.6780	0.5447	0.5170
cpt-16-2K-1.58-10K	0.4526	0.4464	0.6900	0.3375	0.8105	0.2700	0.6621	0.4874	0.6760	0.5539	0.5146
cpt-16-2K-1.58-10K-nowarm	0.4456	0.4107	0.6300	0.3342	0.8134	0.2700	0.6480	0.5487	0.6760	0.6101	0.4988
cpt-16-2K-1.58-10K-opt	0.4632	0.5179	0.6300	0.3376	0.7951	0.2620	0.6420	0.4946	0.6830	0.4989	0.5264
cpt-16-4K-1.58-10K	0.4544	0.4286	0.6500	0.3328	0.8122	0.2680	0.6458	0.5235	0.6520	0.5229	0.5146
cpt-16-4K-1.58-10K-nowarm	0.4491	0.4286	0.6300	0.3329	0.8122	0.2560	0.6415	0.5668	0.6660	0.5745	0.5107
cpt-16-4K-1.58-10K-opt	0.4439	0.4107	0.6200	0.3339	0.8122	0.2780	0.6491	0.5451	0.6600	0.5092	0.4996
cpt-16-6K-1.58-10K	0.4351	0.3929	0.6000	0.3262	0.8122	0.2560	0.6366	0.5451	0.6790	0.5791	0.4838
cpt-16-6K-1.58-10K-nowarm	0.4491	0.4286	0.6100	0.3240	0.8122	0.2640	0.6398	0.5487	0.6700	0.5183	0.4988
cpt-16-6K-1.58-10K-opt	0.4333	0.3750	0.6100	0.3291	0.8122	0.2760	0.6360	0.5090	0.6810	0.5493	0.5107



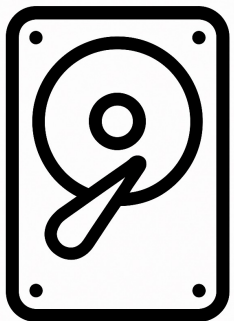
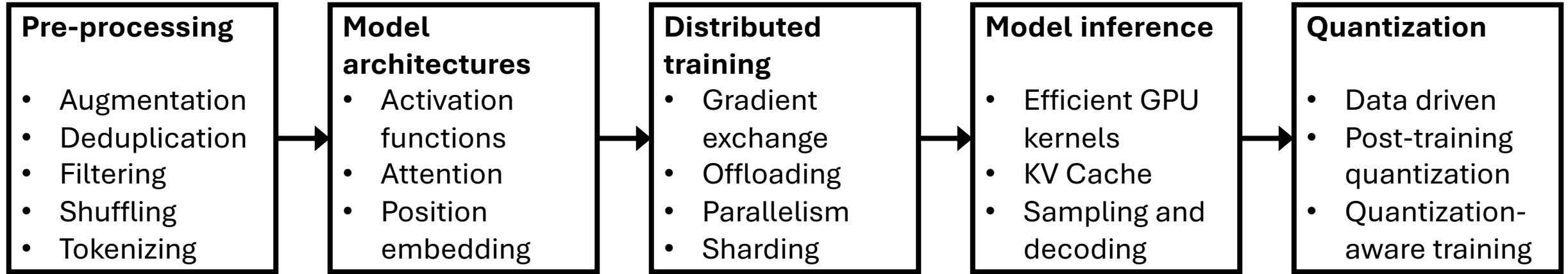
From raw data to deployed model

Quantization

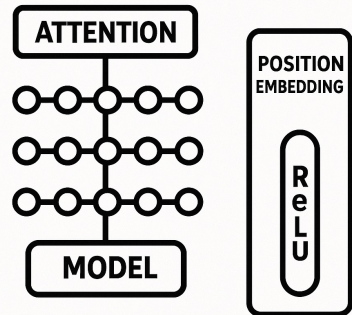
- Data driven
- Post-training quantization
- Quantization-aware training



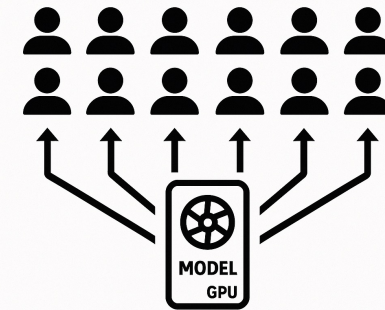
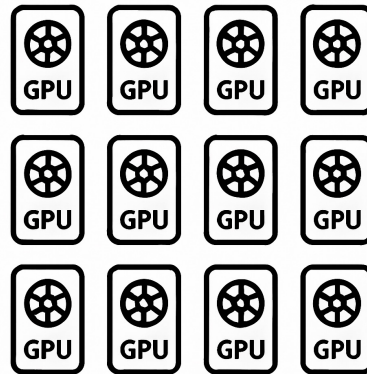
From raw data to deployed model



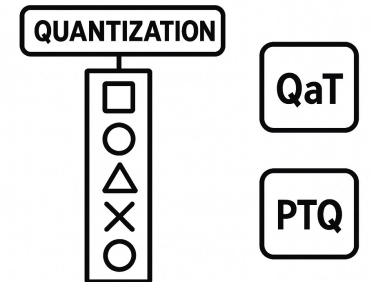
**TEXT
DATA
8TB**



MODEL ARCHITECTURES



THOUSANDS OF USERS



QUANTIZATION

Take-home messages

- Efficiency is not a side quest – it is the only path to democratized AI.
- Classical computer science & mathematics hold great potential.
- Instead of marginal accuracy gains, focus on radical efficiency gains!



Teamwork is dreamwork!



**Andrea
Blasi
Nuñez**



**Jacob
Nielsen**



**Mogens
Henrik
From**



**Lukas
Galke**



**Peter
Schneider-Kamp**